

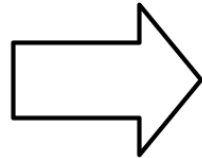
Reshaping data

Logistics and reminders

- HW 1 due tonight
- Reminder: bank of 5 extension days, can use 1 or 2 days on an assignment
- Reminder: annotated course notes and activity solutions posted on course website
- HW 2 released, due next Friday

Last time: `pivot_longer`

id	bp1	bp2
A	100	120
B	140	115
C	120	125



id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

```
1 df |>
2   pivot_longer(
3     cols = bp1:bp2,
4     names_to = "measurement",
5     values_to = "value"
6   )
```

Warmup activity

course website

Work on the activity (~~handout~~) with a neighbor, then we will discuss as a class

Warmup

```
1 df_3
2
3 df_3 |>
4   pivot_longer(cols = -id,
5                 names_to = c(".value", "stage"),
6                 names_sep = "_")
```

	id	bp_1	bp_2	hr_1	hr_2
1	1	100	120	60	77
2	2	120	115	75	81
3	3	125	130	80	93

	id	stage	bp	hr
1	1	1	100	60
2	1	2	120	77
3	2	1	120	75
4	2	2	115	81
5	3	1	125	80
6	3	2	130	93

placeholders for
arbitrary column names
First part stays as
column names

second part of
column names
move to a new
column (stage)

What is `names_to = c(".value", "stage")` doing?

What we can do with the reshaped data

```
# A tibble: 6 × 4
```

	id	stage	bp	hr
	<dbl>	<chr>	<dbl>	<dbl>
1	1	1	100	60
2	1	2	120	77
3	2	1	120	75
4	2	2	115	81
5	3	1	125	80
6	3	2	130	93

Call:

```
lm(formula = bp ~ hr + stage, data =  
df3_new)
```

Coefficients:

(Intercept)	hr	stage2
38.022	1.074	-6.223

```
1 lm(bp ~ hr + stage,  
2   data = df3_new)
```

Going the other way

Data on air quality in two locations (BETR801, London Westminster) on different days:

```
1 air_quality
```

```
# A tibble: 1,825 × 3
```

	date.utc	location	value
	<dtm>	<chr>	<dbl>
1	2019-06-18 06:00:00	BETR801	18
2	2019-06-17 08:00:00	BETR801	6.5
3	2019-06-17 07:00:00	BETR801	18.5
4	2019-06-17 06:00:00	BETR801	16
5	2019-06-17 05:00:00	BETR801	7.5
6	2019-06-17 04:00:00	BETR801	7.5
7	2019-06-17 03:00:00	BETR801	7
8	2019-06-17 02:00:00	BETR801	7
9	2019-06-17 01:00:00	BETR801	8
10	2019-06-16 01:00:00	BETR801	15

BETR 801

London Westminster

What if I want a separate column for each location?

pivot_wider

A tibble: 3 × 3

	date.utc	location	value
	<dtm>	<chr>	<dbl>
1	2019-06-18 06:00:00	BETR801	18
2	2019-06-17 08:00:00	BETR801	6.5
3	2019-06-17 07:00:00	BETR801	18.5

```
1 air_quality |>
2   pivot_wider(id_cols = ..., < columns to keep (date.utc)
3               names_from = ..., < new column names (location)
4               values_from = ...) < values to fill in (value)
```

date.utc	BETR801	London
	18	
	6.5	
	18.5	
	.	
	.	

pivot_wider

```
# A tibble: 3 × 3
```

	date.utc	location	value
	<dtm>	<chr>	<dbl>
1	2019-06-18 06:00:00	BETR801	18
2	2019-06-17 08:00:00	BETR801	6.5
3	2019-06-17 07:00:00	BETR801	18.5

```
1 air_quality |>
2   pivot_wider(id_cols = date.utc,
3               names_from = location,
4               values_from = value)
```

```
# A tibble: 1,670 × 3
```

	date.utc	BETR801	`London Westminster`
	<dtm>	<dbl>	<dbl>
1	2019-06-18 06:00:00	18	7
2	2019-06-17 08:00:00	6.5	6
3	2019-06-17 07:00:00	18.5	6
4	2019-06-17 06:00:00	16	6
5	2019-06-17 05:00:00	7.5	6
6	2019-06-17 04:00:00	7.5	6
7	2019-06-17 03:00:00	7	6

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

For next time, read:

- Chapter 5 in *Modern Data Science with R*